

1. Administrative & Study Identification

Full Study Title: A Study of Zilovetamab Vedotin (MK-2140/VLS-101) in Participants With Solid Tumors (MK-2140-002) **Short Title/Acronym:** Phase 2 Study of VLS-101 in Solid Tumors **Protocol Number:** MK-2140-002 (VLS-101-0003) **NCT Number:** NCT04504916 **Sponsor Name:** Merck Sharp & Dohme LLC (MSD) **Investigational Product:** Zilovetamab Vedotin (MK-2140 / VLS-101) **Phase of Development:** Phase II **Medical Monitor:** MSD Clinical Operations Lead

2. Study Synopsis & Rationale

2.1 Study Objective Primary Objective: To evaluate the objective response rate (ORR) of zilovetamab vedotin in participants with metastatic solid tumors per RECIST version 1.1 by blinded independent central review (BICR). **Secondary Objectives:** To evaluate progression-free survival (PFS), overall survival (OS), duration of response (DOR), time to response (TTR), and to assess the safety and tolerability profile.

2.2 Background & Rationale Zilovetamab vedotin is an antibody-drug conjugate targeting the receptor tyrosine kinase-like orphan receptor 1 (ROR1). While it has demonstrated manageable safety and promising antitumor activity in participants with relapsed or refractory non-Hodgkin lymphomas, this study aims to evaluate its clinical efficacy, safety, and pharmacokinetics in previously treated metastatic solid tumors.

3. Study Design & Methodology

3.1 Design Framework Study Type: Interventional **Control Type:** Single-arm **Blinding:** Open-label **Randomization:** Nonrandomized

3.2 Planned Enrollment & Duration Target \$N\$: 102 evaluable participants **Number of Sites:** Multiple global study sites **Estimated Study Start:** 14-Feb-2022 **Actual Primary Completion:** 12-Jun-2023

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Has a confirmed diagnosis of solid tumor for one of the following types of cancer: previously treated triple-negative breast cancer (TNBC), non-TNBC HER2-negative breast cancer, non-squamous non-small-cell lung cancer

(NSCLC), gastric cancer, pancreatic cancer, or platinum-resistant ovarian cancer. 2. Has metastatic disease that has progressed during or following previous treatment appropriate for the disease type. 3. Presence of radiographically measurable disease and willingness to provide tumor tissue.

4.2 Exclusion Criteria 1. Has peripheral neuropathy of Grade > 1. 2. Has a malignancy involving the central nervous system, or another major cancer. 3. Is pregnant or breastfeeding, has an uncontrolled ongoing infection, or has had major surgery within 4 weeks before the start of study therapy.

5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency: Participants receive zilovertamab vedotin either ≤ 2.5 mg/kg once every 3 weeks (Q1/3W) on Day 1, or < 1.75 mg/kg twice every 3 weeks (Q2/3W) on Day 1 and Day 8. **Route of Administration:** Intravenous (IV) infusion. **Duration of Treatment:** Repeated 21-day cycles until progressive disease or treatment discontinuation.

5.2 Concomitant & Prohibited Therapy Permitted: Standard supportive oncology care. **Prohibited:** Concurrent participation in another therapeutic or imaging clinical trial, or use of a prior MMAE-containing drug.

6. Study Timeline & Visit Schedule

The study involves a structured visit schedule starting with a Screening phase (Day -28 to 0) for obtaining Informed Consent, performing Tumor Imaging, recording Medical History, and assessing Organ Function Labs. At Baseline (Day 1 of Cycle 1), participants receive their First IV Infusion and undergo Baseline AE/Safety Labs. During the Interim phase (Every 21 Days), participants undergo Safety Labs, Efficacy Assessment (RECIST 1.1), and receive their IV Infusion on Day 1 (and Day 8 for the Q2/3W cohort). The End of Study (triggered by EOT / Progression) includes Final Vital Signs, an Exit Interview, IP Accountability, and ongoing Follow-up for PFS/OS.

Visit ID	Window (Days)	Key Activities/Procedures
Screening	Day -28 to 0	Informed Consent, Tumor Imaging, Medical History, Organ Function Labs
Baseline	Day 1 of Cycle 1	First IV Infusion, Baseline AE/Safety Labs
Interim	Every 21 Days	IV Infusion on Day 1 (and Day 8 for Q2/3W cohort), Safety Labs, Efficacy Assessment (RECIST 1.1)
End of Study	EOT / Progression	Final Vital Signs, Exit Interview, IP Accountability, Follow-up for PFS/OS

7. Outcome Measures (Endpoints)

7.1 Primary Endpoint Definition: Objective Response Rate (ORR), defined as the proportion of participants with a confirmed complete or partial response. **Measurement Method:** Evaluated per RECIST Version 1.1 by Blinded Independent Central Review (BICR).

7.2 Secondary & Exploratory Endpoints * Progression-Free Survival (PFS) * Overall Survival (OS) * Duration of Response (DOR) and Time to Response (TTR) * Correlation of ROR1 protein expression with clinical outcomes

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Routine evaluation of treatment-related adverse events (most commonly fatigue, nausea, and peripheral neuropathy) throughout the dosing cycles. **Serious Adverse Events (SAEs):** Reported per standard protocol within 24 hours. AEs leading to dose interruptions/reductions or permanent discontinuation are closely tracked. **Data Monitoring Committee (DMC):** Internal safety review handled by MSD to monitor dose-limiting toxicities.

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: Intent-to-Treat (ITT) population for primary efficacy; Safety Set for all participants receiving ≥ 1 dose of study drug. **Sample Size Justification:** Designed to evaluate a null hypothesis that the ORR is $\leq 5\%$ against the alternative hypothesis that it is $\geq 20\%$. **Handling of Missing Data:** Standard oncology imputation; patients without evaluable post-baseline scans are typically treated as non-responders for ORR.

10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki. **Monitoring Plan:** Standard monitoring of eCRFs and investigator site files to ensure protocol adherence and patient safety. **Audit Trail:** Efficacy evaluations rely on BICR to eliminate investigator bias, combined with strict eCRF locking.

11. Study Identifiers & Governance

Primary ID: MK-2140-002 **Registry Number:** NCT04504916 **Sponsor/Lead Organization:** Merck Sharp & Dohme LLC **Study Title:** Phase 2 Study of VLS-101 in Solid Tumors **Official Regulatory Title:** A Study of

12. Clinical Framework (Oncology Specific)

Primary Condition: Metastatic Solid Tumors **Diagnosis Threshold:** Radiographically measurable metastatic disease progressing on prior standard therapy **Medical Methodology:** Antibody-Drug Conjugate (ADC) targeting ROR1 **Optimization Target:** Maximize Objective Response Rate (ORR) while managing MMAE-related toxicities

13. Study Procedures & Methodology

Management Pathway: Intravenous antineoplastic therapy protocol **Education Protocol (HCP):** Site initiation and training on grading/managing peripheral neuropathy and infusion reactions **Education Protocol (Patient):** Informed consent, symptom tracking, and self-reporting of neuropathy symptoms **Quality Audit Mechanism:** Blinded Independent Central Review (BICR) of all tumor imaging

14. Observation & Follow-Up Schedule

Total Duration: Continuous until disease progression or unacceptable toxicity **Onsite Visit Cadence:** Every 3 weeks for Q1/3W, or Days 1 and 8 of a 21-day cycle for Q2/3W **Remote Monitoring Frequency:** Between cycles as needed for AE follow-up **Checklist Review Interval:** Prior to the start of each 21-day cycle

15. Sign-off & Approval

The report is formally approved by the Principal Investigator (Jun Zhang, MD, PhD), the MSD Clinical Operations Lead, and the MSD Lead Statistician, with all signatures dated 03-Apr-2026.

Role	Name	Signature	Date	:---	:---	:---	:---
Principal Investigator	Jun Zhang, MD, PhD	____	03-Apr-2026	Clinical Operations Lead	MSD Clinical Operations Lead	____	03-Apr-2026
Lead Statistician	____	03-Apr-2026		Lead Statistician	MSD Lead Statistician	____	03-Apr-2026